

Physics-model-free heat-electricity energy management of multiple microgrids based on surrogate model-enabled multi-agent deep reinforcement learning

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HIGHLIGHTS

- The proposed control framework mitigates assumptions about accurate physical models.
- A novel exploration mechanism incorporates historical information representation.
- Each MG controller can manage the energy of MG in a decentralized manner.

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ABSTRACT

Reliability and cost-effectiveness in the operation of the multiple microgrid (MMG) system depend on the skillful management of its energy resources. Traditional energy management approaches are physics-model-based, which rely on the precise system parameters (e.g., line parameters) of the electricity and heat network. It is difficult to establish the precise system parameters in practice because they depend on a variety of factors. In this context, this paper proposes a physics-model-free control framework for the energy management of MMGs with heat-electricity energy, consisting of proposed surrogate model and multi-agent deep reinforcement learning (MADRL) approach. An important step is to use historical data to train a surrogate model in supervised manner that can imitate the realistic power and thermal flow calculations. Meanwhile, the energy management problem is reformulated as a Markov game. It is solved by the proposed MADRL-based approach by modeling each MG controller as an agent with a specific objective. The historical trajectories representation, parameter space technology, and deep dense architecture in reinforcement learning are introduced in MADRL to overcome the negative impact brought by the time series data from the input state on the decision-making process and construct an efficient exploration mechanism to overcome inefficient optimization of the MMG system in a multi-agent setting. During training period of MADRL, the trained surrogate models integrate into the environment of the MADRL, which can develop optimal energy management strategy based on the continuous interaction with the surrogate models. The proposed surrogate model enabled MADRL approach can reduce the reliance on precise physical systems and prevent having an impact on the real system while being trained involving trial and error process. Simulation results demonstrate the effectiveness of the proposed control framework.

1. Introduction

A vigorous effort has been made globally to develop renewable energy sources such as wind and solar power to reduce dependence on fossil fuels and greenhouse gas emissions [1,2]. In this context, microgrids (MG) provide an effective way of utilizing distributed renewable

energy by deploying distributed generators, distributed energy storage, and local demands in a decentralized manner [3,4]. The present day multi-energy MGs are capable of improving holistic energy utilization efficiency through the integration of heterogeneous energy carriers such as electrical and thermal energy, despite their typically small size. Due to the capacity limitations of an individual MG, there are places where

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multiple nearby MGs can exchange energy to reduce total costs and bolster the reliability of the power supply [5]. Therefore, the multiple MGs (MMGs) with combined heat and power (CHP) can be regarded as a novel concept with both high energy utilization efficiency and high energy allocation potential. To exploit the advantages of MMG with CHP, it is necessary to develop energy management for it. Developments in energy management for the MMG with CHP system typically classified into two types: centralized and distributed or decentralized approach.

Ref. [6] uses particle swarm optimization to centrally schedule MMGs with the lowest operational costs. However, owing to the high levels of real-time necessary for MMG control and the complexity of energy management of MMGs, the [6] cannot accomplish the intended results. In view of this, [7] presents an imperialist competitive-based approach for addressing the energy management issue of MMG with CHP system, which has quicker convergence speed and stronger global convergence than [6] to try to satisfy high real-time control requirements and achieve better control performance. Centralized control is required to process massive volumes of information at a single location simultaneously. Because of this characteristic, centralized control cannot provide the plug-and-play functionality that is essential for the MMG system; centralized control is better suited for stand-alone power systems that maintain critical supply and demand balances in an infrastructure that changes slowly [8]. Additionally, centralized control needs to collect the information of all the MGs. Distinct MGs may be owned by different parties, and the scheduling of distributed generations inside each MG may be governed by their own policies and economic rules [9]. This may restrict the exchange of information between MGs, for concerns of privacy and security.

In distribution-based and decentralization-based approaches, the whole optimization problem is divided into a number of smaller sub-problems that can be solved in a distributed manner, potentially overcoming the drawbacks of the centralized approach to some extent. For this reason, distributed and decentralized control manner is a trend for the energy management of MMGs [10]. To quantify the benefits of proactive resource trading among MMGs, [11] proposes a distributed alternating direction method of multipliers (ADMM) to optimize synergistic MMG with CHP activities, which can discover the optimal solution in less iterations. Further, [12] implements the primal dual multiplier method to the energy management of MMG in distributed manner, which delivers better results in terms of optimized time spent and solution accuracy than [11], obtaining the required results. Therein, information is shared among MGs to enhance the coordination. However, such a procedure would increase communication costs, and for information sharing reasons, the privacy cannot be protected well. To fill the gaps, decentralized control manner is the desired way. The decentralized strategy does not necessitate the establishment of communication links between controllers. Some of the distributed approach's characteristics, as well as the fact that the control is conducted using just local information, are kept in decentralized control manner. In [13], a robust decentralized optimization is presented to manage the source in MMG integrated with CHP. To improve the energy management of MMG containing CHP, a decentralized bi-level stochastic optimization technique is presented in [14]. Distributed energy resources are subject to rapid fluctuations. For instance, due to cloud dynamics, the photovoltaic (PV) cell output may fluctuate quickly in a matter of seconds. To cope with the fast-changing outputs of PVs, more frequent operations of controlled devices are necessary under this circumstance. Though the above studies achieve some good results, they may hard to respond to the last optimization requirement when DERs change dramatically in a short period of time.

Recently, the emergence of deep reinforcement learning (DRL) is expected to provide a way to fill the gaps. In the case of other domains, the use of DRL has proven to be a viable way to coordinate the operation of different system resources in the most efficient manner possible while taking into account their specific characteristics and uncertainties.

Unlike optimization-based approaches, the obtained policy from DRL can be used to make end-to-end complex decisions by generating the desired action instantly (e.g., within several milliseconds) based on the state of the system right after the learning process [15]. A DRL system with this characteristic is well suited for applications that involve the decision-making process in real-time. Through the integration of DRL's expertise into power system decision-making and control, a flexible tool can be developed to overcome the limitations of traditional model-based control approaches and to allow fast and effective decision making in highly complex situations. In addition, the use of model-free DRL does not have to be detrimental to the modeling efforts that are already being undertaken; rather, it may be regarded as a complementary role in order to achieve better degrees of adaptability and flexibility. In this way, it has the potential to be used as a strategic tool that helps people make decisions that are adaptive, effective, flexible, quick, and forward-looking. In today's constantly changing world, this is a great way to meet the needs of a rapidly changing energy landscape [16]. Despite the continuous advancement of DRL application research in energy management, current approaches primarily focus on single-agent DRL-based control strategies, which limit the application of single-agent DRL to specific small-scale scenarios when the operation state dimensions of emerging power and energy networks are becoming more and more multidimensional. Problems requiring complex coordination among agents can be poorly solved using single-agent DRL approaches [17]. In this regard, multi-agent DRL (MADRL) has emerged as a promising active research area that merges both the advantages of multi-agent systems and data-driven approaches, and it has the ability to efficiently handle decision-making problems in an environment with multiple agents, and as such is receiving a lot of attention.

A multi-agent Q-learning based residential multiple energy MG (MEMG) energy management approach is proposed in the [18]. Although the simulation results demonstrate its effectiveness, it cannot solve the continuous state and action space. The fuzzy Q-learning based approach, which can overcome the "curse of dimensionality" problem that traditional Q-learning may encounter, is proposed in [19] to construct a cooperative multi-agent system for managing the energy of the MG. The experimental findings demonstrate the ability of the multi-agent system to ensure electricity supply and improve the constancy of the MG. Nevertheless, fuzzy-Q learning may have sluggish convergence and require manual extraction of fuzzy features, making it susceptible to local optimum. To lower the computational cost of RL, a sizable research effort is still required. To this end, [20] proposes a function approximation for Q-learning to leverage a diffusion strategy to coordinate the actions among multiple agents to manage the energy in the microgrid. Simulation results demonstrate its advantages compared to fuzzy-Q learning. However, it only utilizes linear mapping to approximate the Q-value table, which makes the approximate results insufficient to fully characterize the implicit information in the Q-value table. In order to better approximate the Q-value table, [21] proposes a multi-agent double deep Q network (MADDQN)-based approach for the MG energy management. Although the simulation results demonstrate its effectiveness, each agent needs to collect other agents' information to make decision. Taking this approach may carry both the risk of privacy leakage and the inevitable cost of communication as well. To avoid these drawbacks to a certain extent, [22] proposes a MADDQN-based approach that uses the sharing policy network between agents to ensure that local information can be utilized to obtain the decision in the energy management of power and transportation coupling networks. However, the parameters of the sharing policy network need to broadcast to all local agents through communication equipment. Due to the huge number of parameters in neural networks, broadcasting parameter information may burden the communication bandwidth, which may result in communication delays and data loss. Therefore, this approach is not an ideal choice.

The approaches that follow the centralized training and decentralized execution manner are good choices to avoid privacy leakages and

reduce communication costs. Specifically, the actor-critic architecture provides the critic network with rich extra information to stimulate the actor network's improvement. This technique allows each agent's actor network to maintain a coordinated interaction with others even when it is just dependent on local information at the execution stage [17]. By utilizing this type of approach, MMG's energy management can be accomplished in a totally decentralized manner. This results in a fresh guideline for research on MMG energy management, taking into account single-point failure avoidance, privacy protection, minimizing communication costs associated with communication device deployment, and real-time responsiveness to optimization demand. Therefore, in recent years, there have been many literatures on this topic. In [23], multi-agent asynchronous advantage actor-critic (MAA3C)-based approach is used to build an energy scheduling model for the MG-powered multi-access edge computing network. A multi-agent trust region policy optimization (MATRPO)-enabled MMG system is developed in [24] to minimize total costs and improve grid reliability. Ref. [25] proposes a PPO with parameter sharing-based MADRL approach for the real-time automatic routing and scheduling problem of multiple electric vehicles in the coupled power-transportation network. In [26], a multi-agent proximal policy optimization (MAPPO)-based approach is proposed for the objective of MEMG system resilience enhancement.

The above mentioned actor-critic MADRL approaches belong to on-policy. Compared to on-policy MADRL approaches, the data used for the training of off-policy MADRL approaches can be from self-generated and external, with more abundant means of utilizing historical data. Therefore, the latter have higher data utilization rates than the former, and their use in optimizing energy management is also more common. A multi-agent deep deterministic policy gradient (MADDPG)-based energy management is proposed in [27] to efficiently coordinate dispatchable units to meet the load and reduce operating costs in MG. In [28], the MADDPG-based approach is proposed to manage the energy of multiple energy hubs (EHs) with multiple energies. In [29], a MADDPG-based approach is proposed to reduce multiple MGs own energy bills and cut down the load on the main utility grid. In [30], multi-agent twin delayed deep deterministic policy gradient (MATD3)-based approach is proposed to optimize energy trading of MMG with CHP. Ref. [31] proposes a MATD3 with attention mechanism-based approach to manage the multiple energies in the MMG system. In order to manage the multiple energies involved in the MMG system and reduce energy costs and carbon emissions, [32] proposes a MATD3-based approach. As a means of controlling MMG's economic frequency, a multi-agent soft actor critic (MASAC)-based approach is presented in [33]. In [34], a MASAC-based approach is proposed to manage the energy of MMG with CHP. In [35], the multi-actor-attention-critic (MAAC)-based approach is proposed to manage the energy of multiple EHs to achieve the minimization of energy costs under different demands. A MAAC-based approach is proposed in [36] to manage energy in the EHs.

To highlight the contributions of this paper, existing literature associated with the energy management problems of MG-like has been systematically organized in Table 1.

Firstly, compared with [18–22], this paper utilizes MADRL with off-policy actor-critic framework to manage the energy of MMG system. The MADRL with actor-critic framework features centralized training and decentralized execution. This allows the actor network of each MG controller to obtain a coordinated schedule plan even when only local information is available at the time of execution.

Secondly, despite the advantages indicated, Refs. [23–36] have certain drawbacks: since the reward function is calculated using the power and thermal flows, the premise that one has accurate knowledge of the parameters and topology of the electricity and heat networks remains. However, it is hard to make accurate physical models of real-world systems' electrical and thermal systems. For electrical system, it is sometimes difficult for operators to be aware of the precision line parameters and topology [37,38]. Additionally, it is challenging to accurately estimate these characteristics of electrical system because it

Table 1

Summary of Existing Literature associated with the MG-like Energy Management.

Paper	Research object	Algorithm type	Basic Algorithm	Independent on mathematical model	Effective exploration mechanism
[18]	MEMG	Value-based	MAQ-learning	×	×
[19]	MG	Value-based	Fuzzy Q-learning	×	×
[20]	MG	Value-based	MAQ-learning	×	×
[21]	MG	Value-based	MADDQN	×	×
[22]	MG	Value-based	MADDQN	×	×
[23]	MG	Actor-critic (on-policy)	MAA3C	×	×
[24]	MG	Actor-critic (on-policy)	MATRPO	×	×
[25]	MG	Actor-critic (on-policy)	PPO	×	×
[26]	MEMG	Actor-critic (on-policy)	MAPPO	×	×
[27]	MG	Actor-critic (off-policy)	MADDPG	×	×
[28]	EH	Actor-critic (off-policy)	MADDPG	×	×
[29]	MG	Actor-critic (off-policy)	MADDPG	×	×
[30]	MEMG	Actor-critic (off-policy)	MATD3	×	×
[31]	MEMG	Actor-critic (off-policy)	MATD3	×	×
[32]	MEMG	Actor-critic (off-policy)	MATD3	×	×
[33]	MG	Actor-critic (off-policy)	MASAC	×	×
[34]	MEMG	Actor-critic (off-policy)	MASAC	×	×
[35]	EH	Actor-critic (off-policy)	MAAC	×	×
[36]	EH	Actor-critic (off-policy)	MAAC	×	×
This	MEMG	Actor-critic (off-policy)	MAAC	✓	✓

either necessitates a large number of time-stamped measurements or the complete measured values of the phasor measurement unit. The former may take a lengthy time, whereas the latter is seldom used in reality. For thermal systems, accuracy in pipe parameters and topology is also required for thermal flow computation, similar to electrical systems. Because thermal systems have things like complicated response mechanisms, high non-linearity, significant lag, etc., it is hard to get the exact parameters of the pipes in the system. In addition, because the consumers are typically widely distributed in the heat network, the topology of the secondary pipelines is rather complex [39]. These reasons bring challenges for accurately establishing the mathematical model of the real-world heat network. Compared with [23–36], the proposed

control framework include the surrogate model to simulate the power flow and thermal flow calculation processes. Given the limited historical data available to build the surrogate model for the MMG system, unlike normal deep learning technologies, the proposed surrogate model with few-shot learning characteristic is capable of constructing ideal mapping relationships between input data and output power and thermal flow results utilizing relatively few data points. This eliminates the need for precise model parameters of electricity network and heat network, which are notoriously hard to come by and are subject to a wide range of practical errors.

Finally, most studies have focused on altering the architecture of critic networks in order to enhance coordination among the agents, without delving into the potential of actor networks to improve the energy management of MMG [23–36]. Unlike individual MG optimization, the action space in the multi-agent situation grows exponentially as the number of MG rises. A broader action space would make it more difficult for agents to uncover excellent control strategy by naive trial and error, especially in the coordinated scenario [40], which readily leads to premature convergence to the local optimal solution. These issues may restrict the use of MADRL to energy management of MMG. To this end, a novel exploration mechanism is introduced in the proposed control framework to avoid convergence prematurely to a local optimum due to the huge action space in energy management of the MMG system optimized by a multi-agent control framework. As opposed to traditional action space noise (ASN) that is added directly to the action [23–36], the proposed exploration mechanism is deployed as follow: the gated recurrent network (GRU) is firstly utilized to get a representation of historical information for the actor network to overcome the negative impact brought by the time series data (TSD) from the input state on the decision-making process, then, the parameter space noise (PSN) is deployed in the parameters of actor network to increase the scope of decision-making space exploration. In the meantime, the deep dense architecture in reinforcement learning (D2RL) is deployed in the actor network to mitigate the training instability and inefficiency caused by noise.

In this paper, a physics-model-free control framework for the energy management of MMGs with heat-electricity energy is developed to minimize the operation cost while satisfying the voltage limitation, which includes the proposed surrogate model and the MADRL approach. The main contributions and salient features of this paper are listed as follows:

- 1) To simulate the power flow and thermal flow calculation processes, two categories of surrogate models are trained using sparse variational Gaussian processes (SVGP). Using Bayesian theory-based technique, only a small amount of historical data is needed to successfully detect the mapping connection between input and desired output.
- 2) A novel exploration mechanism incorporating historical information representation is proposed to embed into the actor network. The GRU is utilized to get a representation of historical information for the actor network to overcome the negative impact brought by the TSD from the input state on the decision-making process. For multi-agent scenarios, the PSN is utilized to construct effective exploration mechanisms in order to avoid premature convergence to a local optimum. In addition, the D2RL is deployed in actor network to improve the training instability and inefficiency caused by the introduction of noise. PSN and D2RL are complementary because PSN acts at the parameter level while D2RL establishes a more complicated and effective parametric link relationship, thereby increasing the exploratory action potential while ensuring training stability and efficiency.
- 3) The proposed control framework does not necessitate real-time optimization of the complex MMG energy management problem, as is the case with model-based approaches. The proposed control framework enables MG controllers to construct decision-making

functions offline and then deploy them online, allowing for fully decentralized selection of the optimum energy management based on the most up-to-date system data.

The rest of this paper is organized as follows. Section II introduces the mathematical formulations of MMG system and the optimization problem. In Section III, the optimization problem is reformulated as Markov games and an introduction to the proposed control framework is provided. Section IV analyses the numerical simulation results. The conclusion of the paper is summarized in Section V.

2. System modeling descriptions

2.1. Interconnected MMG with CHP model

An illustration of a grid interconnected MMG with CHP system is shown in Fig. 1. The number of MG is denoted as $g \in \{1, 2, 3, \dots, G\}$. The g -th MG is made up of a heat and electricity network. The heat network is made up of CHP, power to heat (PtH) unit (such as heat pump), and thermal load. The electricity network is made up of diesel generator (DEG), renewable energy (such as solar and wind energy), energy storage system (ESS), and electrical load. The individual MGs are integrated into an interconnected MMG system by the power line (black line) and heat pipe (red line), allowing for the exchange of electrical and thermal energy between the different MGs. Typically, there are circuit breakers as ON or OFF switches in the MMG system. This paper assumes circuit breakers are always ON to guarantee energy exchange.

2.2. Electricity network modelling

Power flow calculation results are the basis for analyzing network planning and electricity supply scheme feasibility, safety, reliability, and economy. Power flow analysis plays an imperative role in the operation mode and design of electricity networks.

The power flow constraints are as follows:

$$P_{i,t} = V_{i,t} \sum_{j=1}^J V_{j,t} (G_{ij,t} \cos \theta_{ij,t} + B_{ij,t} \sin \theta_{ij,t}) \quad (1)$$

$$Q_{i,t} = V_{i,t} \sum_{j=1}^J V_{j,t} (G_{ij,t} \sin \theta_{ij,t} - B_{ij,t} \cos \theta_{ij,t}) \quad (2)$$

where $P_{i,t}$ is active power injected into node i at time t ; $Q_{i,t}$ represents the reactive power injected into node i ; J denotes the total number of nodes; $V_{i,t}$ denotes the voltage amplitude of node i ; $G_{ij,t}$ and $B_{ij,t}$ are the real and imaginary part of admittance element between node i and j ; $\theta_{ij,t}$ is the voltage phase difference between node i and j .

The following restrictions apply when calculating power flow:

- Node voltage amplitude: The node voltage should be in the allowed range in order to comply with the requirements for electricity quality and electrical supply safety.

$$V_{\min} \leq V_{i,t} \leq V_{\max} \quad (3)$$

where $V_{\min}$ and $V_{\max}$ are the lower and upper limits of node i .

- Generation node: The active and reactive power at the generation nodes must be within a safe and reasonable range to ensure the stability of the system. In addition, some nodes with distributed generation must satisfy the working constraints (e.g. DEG, etc.).

$$0 \leq P_{i,t}^{PV} \leq P_{i,\max}^{PV} \quad (4)$$

$$0 \leq P_{i,t}^{WT} \leq P_{i,\max}^{WT} \quad (5)$$

$$0 \leq P_{i,t}^{PtH} \leq P_{i,\max}^{PtH} \quad (6)$$

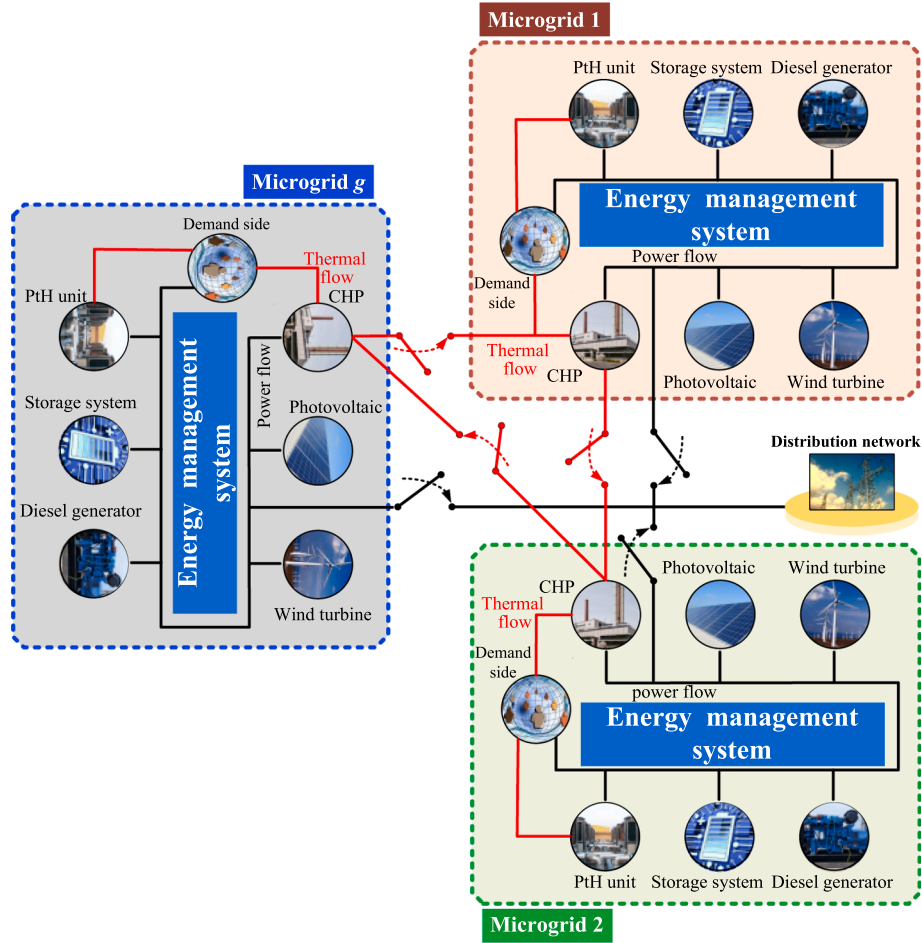


Fig. 1. Illustration of a grid interconnected MMG with CHP system.

$$P_{i,t}^{DEG} - P_{i,t-\Delta t}^{DEG} \leq R_{up}^{DEG} \Delta t \cdot s_{i,t-\Delta t}^{DEG} + P_{imin}^{DEG} (s_{i,t}^{DEG} - s_{i,t-\Delta t}^{DEG}) + P_{imax}^{DEG} (1 - s_{i,t}^{DEG}) \quad (7)$$

$$P_{i,t-\Delta t}^{DEG} - P_{i,t}^{DEG} \leq R_{dn}^{DEG} \Delta t \cdot s_{i,t}^{DEG} + P_{imin}^{DEG} (s_{i,t-\Delta t}^{DEG} - s_{i,t}^{DEG}) + P_{imax}^{DEG} (1 - s_{i,t-\Delta t}^{DEG}) \quad (8)$$

$$-P_{imax}^{ESS} \leq P_{i,t}^{ESS} \leq P_{imax}^{ESS} \quad (9)$$

$$SOC_{imin}^{ESS} \leq SOC_{i,t}^{ESS} \leq SOC_{imax}^{ESS} \quad (10)$$

$$Q_{imin}^{PV} \leq Q_{i,t}^{PV} \leq Q_{imax}^{PV} \quad (11)$$

$$Q_{imin}^{WT} \leq Q_{i,t}^{WT} \leq Q_{imax}^{WT} \quad (12)$$

$$Q_{imin}^{DEG} \leq Q_{i,t}^{DEG} \leq Q_{imax}^{DEG} \quad (13)$$

$$Q_{imin}^{ESS} \leq Q_{i,t}^{ESS} \leq Q_{imax}^{ESS} \quad (14)$$

where $P_{i,t}^{PV}$, $P_{i,t}^{WT}$, $P_{i,t}^{PtH}$, $P_{i,t}^{DEG}$ and $P_{i,t}^{ESS}$ are the active powers of PV, wind turbine (WT), PtH, DEG and ESS of node i ; R_{up}^{DEG} and R_{dn}^{DEG} denote the ramp-up and ramp-down limits of DEG; $s_{i,t}^{DEG}$ denotes the binary variable, which represents the working state of DEG at time t (1 denotes ON, 0 represents OFF); P_{imax}^{PV} , P_{imax}^{WT} , P_{imax}^{PtH} , P_{imax}^{DEG} and P_{imax}^{ESS} are the upper limits of active powers of PV, WT, PtH, DEG and ESS of node i ; $SOC_{i,t}^{ESS}$ represents the state-of-charge (SOC) of ESS of node i ; SOC_{imax}^{ESS} is the upper limit of SOC; $Q_{i,t}^{PV}$, $Q_{i,t}^{WT}$, $Q_{i,t}^{DEG}$ and $Q_{i,t}^{ESS}$ are the reactive powers of PV, WT, DEG and ESS at node i ; Q_{imax}^{PV} , Q_{imax}^{WT} , Q_{imax}^{DEG} and Q_{imax}^{ESS} are the maximum reactive

powers of PV, WT, DEG and ESS at node i ; Q_{imin}^{PV} , Q_{imin}^{WT} , Q_{imin}^{DEG} and Q_{imin}^{ESS} are the minimum value of reactive powers.

2.3. Heat network modelling

A heat network typically consists of one source node and multiple intermediate/load nodes linked by primary network and secondary network. The primary network is known as the generation-side network, while the secondary network is known as the consumption-side network. During the thermal energy transfer process, water is used to transport thermal energy from the supply pipes in the primary network to the thermal exchanger in the secondary network, where the thermal energy is distributed to all load nodes causing the transmitted water to undergo a drastic temperature shift. Finally, the thermal circulation is completed when the water travels back through the return pipes [41].

1) Hydraulic model

- Continuity of flow: As the definition, continuous flow is when the mass flow into a node is equal to the mass flow out of the node plus the flow consumed at the node [42]. The matrix A depicts the network incidence of the nodes on the branches. The mass flow vector within a pipe and the mass flow vector through a node that is injected from a source or discharged to a load are defined as the vectors m and m_q , respectively.

$$Am = m_q \quad (15)$$

- Loop pressure equation: The pressure difference in meters caused by pipe friction is referred to as head loss. The loop pressure equation

requires that the total head losses around a closed loop be equal to zero. The following is the loop pressure equation for the hydraulic network [42].

$$\mathbf{B}\mathbf{h}_f = 0 \quad (16)$$

where $\mathbf{B}$ is the loop incidence matrix that relates the loops to the branches, and $\mathbf{h}_f$ is the vector of the head losses.

- Head loss equation: Along each pipe, the relationship between the flow and the head losses is as follows [42]:

$$\mathbf{h}_f = \mathbf{K}\mathbf{m}|\mathbf{m}| \quad (17)$$

where $\mathbf{K}$ is the vector of each pipe's resistance coefficients. $\mathbf{K}$ typically has a strong relationship with a pipe's diameter.

2) Thermal model

- Nodal thermal flow balance: Thermal flow follows the Kirchhoff law, which states that water flows in and out of each node in equal amounts.

$$\sum_{p \in \mathbf{P}_{i, \text{start}}^{s/r}} m_{p,t}^{s/r} = \sum_{p \in \mathbf{P}_{i, \text{end}}^{s/r}} m_{p,t}^{s/r} \quad (18)$$

where $\mathbf{P}_{i, \text{start}}^{s/r}, \mathbf{P}_{i, \text{end}}^{s/r}$ denotes the set of pipes starting and ending at node i in supply/return network, respectively; $m_{p,t}^{s/r}$ represents the mass flow rate of supply/return pipe p at time t .

- Nodal temperature mixing: The nodal temperature is dependent on water mass and temperature, as stated by the first law of thermodynamics. A node's temperature is the same as the temperature of its matching starting node in the same pipe. Mathematically, they can express as:

$$T_{i,t}^{s/r} \cdot \sum_{p \in \mathbf{P}_{i, \text{start}}^{s/r}} m_{p,t}^{s/r} = \sum_{p \in \mathbf{P}_{i, \text{end}}^{s/r}} T_{p, \text{end}, t}^{s/r} \cdot m_{p,t}^{s/r} \quad (19)$$

$$T_{i,t}^{s/r} = T_{p, \text{start}, t}^{s/r} \quad (20)$$

where $T_{i,t}^{s/r}$ denotes temperature of node i of supply/return network at time t ; $T_{p, \text{start}, t}^{s/r}$ and $T_{p, \text{end}, t}^{s/r}$ represent temperature at the start and end of supply/return pipe p at time t .

- Internal temperature range: A pipe's water temperature boundaries for inflow and outflow are as follows:

$$T_{p, \text{start}/\text{end}, t}^{s/r, \min} \leq T_{p, \text{start}/\text{end}, t}^{s/r} \leq T_{p, \text{start}/\text{end}, t}^{s/r, \max} \quad (21)$$

Thermal transmission delay: In contrast to the instantaneous nature of electric energy, there is a noticeable delay in the transmission of heat. The water temperature at the beginning and end of a pipe may not be the same because of the sluggish flow speed of water in the pipeline compared with the electric energy transfer rate. This paper refers to [41] for more specific formulas that can be used to quantify this transmission delay.

Thermal energy balance: There should be a balance between thermal energy production and consumption. The total amount of thermal energy needed in the thermal system is equal to the energy difference between the start of the supply pipe and the end of the return pipe, taking into account that hot and cold water are used to transfer thermal energy between the heat source node and the load node via thermal pipe.

$$H_{p,t} = C_{WH} \cdot (m_{p,t}^s \cdot T_{p, \text{start}, t}^{s/r} - m_{p,t}^r \cdot T_{p, \text{end}, t}^{s/r}) \quad p \in \mathbf{N}_{SN} \quad (22)$$

where $H_{p,t}$ denotes the thermal demand of pipe p at time t ; C_{WH} represents water heat capacity, and $\mathbf{N}_{SN}$ is the set of source nodes.

2.4. Optimization problem

Considering the above models, real-time energy management of MMG systems can be viewed as an optimization problem involving minimizing the following objective function.

$$\min \left\{ \sum_t \left(C_t^{EP} + C_t^{\text{Voltage}} + \sum_g \left(C_t^{\text{DEG}} + C_t^{\text{Gas}} \right) \right) \right\} \quad (23)$$

s.t., Eqs.(1) - (22) and thermal transmission delay. Where T denotes the length of a period; G denotes the number of MGs; C_t^{EP} is the cost of buying electricity from the distribution network (DN); C_t^{Voltage} is the cost of unstable voltage; C_t^{DEG} represents the cost of DEG; C_t^{Gas} denotes the cost of buying gas.

$$C_t^{EP} = P_t^{\text{DN}} \cdot EP_t \cdot \Delta t \quad (24)$$

$$C_{i,t}^{\text{Voltage}} = \begin{cases} 1 - 100 \cdot |1 - v_{i,t}^g|, & v_{i,t}^g \in [0.95, 1.05] \\ -100 \cdot |1 - v_{i,t}^g|, & v_{i,t}^g \in [0.8, 0.95] \cup v_{i,t}^g \in [1.05, 1.25] \\ -1000, & \text{Otherwise.} \end{cases} \quad (25)$$

$$C_t^{\text{Voltage}} = \sum_{i=1}^I C_{i,t}^{\text{Voltage}} \quad (26)$$

$$C_t^{\text{DEG}} = C_t^{\text{Fuel}} + C_t^{\text{Operation}} \quad (27)$$

$$C_t^{\text{Fuel}} = \sum_{i=1}^I s_{i,t}^{\text{DEG}} \cdot \left(a \cdot (P_{i,t}^{\text{DEG}})^2 + b \cdot P_{i,t}^{\text{DEG}} + c \right) \cdot \Delta t \quad (28)$$

$$C_t^{\text{Operation}} = \sum_{i=1}^I C_{i,t}^{\text{start/stop}} \cdot |s_{i,t}^{\text{DEG}} - s_{i,t-\Delta t}^{\text{DEG}}| \quad (29)$$

$$C_t^{\text{Gas}} = \sum_{i=1}^I \frac{P_{i,t}^{\text{CHP}} \cdot \Delta t}{\eta_{\text{CHP}} \cdot L_{\text{cvng}}} \cdot GP_t \quad (30)$$

where P_t^{DN} symbolizes the power energy exchange between MMG system and DN at time t ; $v_{i,t}^g$ represents the node voltage; the MMG system's overall node count is indicated by I ; the fuel cost for the g -th MG's DEG is C_t^{Fuel} , while its operating cost is $C_t^{\text{Operation}}$; a , b , and c are coefficients of the DEG fuel cost function; $C_{i,t}^{\text{start/stop}}$ is the startup or shutdown cost of DEG at node i in g -th MG; $s_{i,t}^{\text{DEG}}$ denotes the working state of DEG; η_{CHP} is the energy generation efficiency of micro turbine; L_{cvng} represents the low calorific value of natural gas, and GP_t denotes the gas price.

3. Proposed physics-model-free framework for energy management of MMG system

This paper proposes a physics-model-free control framework for the energy management of MMGs with heat-electricity energy.

Conventional optimization strategies rely on well-defined electrical and thermal network parameters and topological structure. In actuality, they are often unknown to the operators. In such cases, it is hard to model the premise power and thermal flow model to calculate the realistic flow results (i.e., $P_t^{\text{DN}}, v_{i,t}^g$ and $P_{i,t}^{\text{CHP}}$) in the Eq.(23). To solve this problem, surrogate models is proposed to obtain realistic calculation results for power and thermal flow. Later, the trained surrogate models are integrated into the environment of the proposed MADRL approach to construct the proposed physics-model-free control framework.

3.1. Proposed surrogate model

Given the load requirements of each node, the power introduced by

the renewable energy and CHP and controllers' decision, the electrical surrogate model attempts to estimate the power exchange between MMG system and DN P_t^{DN} , and the voltages of each node of the MMG system $v_{i,t}^g$. Accounting for the losses in the thermal flow and given the heat output of the heat pump, the goal of the thermal surrogate model is to figure out how much heat the CHP needs to produce $P_{i,t}^{g,CHP}$ to meet the heating load. Thermal and electrical surrogate models' mapping correlations are learnt by historical data. Unlike deep learning technologies, the proposed surrogate model can construct ideal mapping relationships between input data and output power and thermal flow results in relatively few data, which has a few-shot learning characteristic, considering the historical data of the MMG system used to build the surrogate model is limited. This eliminates the need for precise model parameters of electricity network and heat network, which are notoriously hard to come by and are subject to a wide range of practical errors.

Assume that there is a relationship in the training data between the input $X \in R^{N \times Q}$ and the output $Y \in R^{N \times 1}$ is subject to the Gaussian Process (GP):

$$Y = f(X) + \varepsilon \quad (31)$$

where $\varepsilon \sim \mathcal{N}(0, \sigma_n^2 I)$ stands for Gaussian white noise, σ_n^2 represents variance; I denotes an identity matrix with appropriate dimensions, and $f(\cdot)$ represents Gaussian process mapping between input and output:

$$f(x) \sim GP(m(x), K(x, x')) \quad (32)$$

where $x, x' \in X$, the mean and covariance matrices of $f(x)$ are represented, respectively, by $m(x)$ and $K(x, x')$. Most commonly, $m(x)$ is assumed to be equal to zero if there are few priors, and elements in the covariance matrix $k(\cdot)$, which is also called the kernel function, can be expressed as follows:

$$k(x, x') = \sigma_k^2 e^{-0.5(x-x')^2} \quad (33)$$

where σ_k denotes covariance.

After knowing m and k , the prior distribution of y can be expressed as:

$$y \sim \mathcal{N}(0, K(X, X) + \sigma_n^2 I) \quad (34)$$

When new input x^* in test data are given, the joint prior distribution of y and f_* is:

$$\begin{bmatrix} y \\ f_* \end{bmatrix} \sim \mathcal{N}\left(0, \begin{bmatrix} K(X, X) + \sigma_n^2 I & K(X, x^*) \\ K(x^*, X) & K(x^*, x^*) \end{bmatrix}\right) \quad (35)$$

Thus, the posterior distribution of predicted value f_* can be obtained by [43]:

$$f_* | X, y, x^* \sim \mathcal{N}(\mu^*, \Sigma^*) \quad (36)$$

where μ^* and Σ^* represent the mean and variance of test data, respectively, and they can be calculated as [43]:

$$\mu^* = K(x^*, X)(K(X, X) + \sigma_n^2 I)^{-1} y \quad (37)$$

$$\begin{aligned} \Sigma^* &= K(x^*, x^*) - \\ &K(x^*, X)(K(X, X) + \sigma_n^2 I)^{-1} K(X, x^*) \end{aligned} \quad (38)$$

When the parameters $\theta = \{\sigma_n, \sigma_k\}$ are known, the mean and variance of test data can be calculated according to Eq. (37) and (38). The parameters θ can be updated by maximizing the negative log-likelihood function $-\log p(Y|X)$ [43].

The calculation complexity of the above mentioned step is $O(N^3)$ where the N is the input $X \in R^{N \times Q}$ or output $Y \in R^{N \times 1}$ data's data size. When dealing with enormous amounts of data, GP may face a significant computational strain. To reduce computational complexity, a

variational inference approach known as SVGP is used, in which M ($M \ll N$) auxiliary input $Z \in R^{M \times Q}$ and corresponding outputs $u \in R^{M \times 1}$ are introduced. After the above improvement, the computational complexity is reduced to $O(NM^2)$. The joint density of y, f and u is:

$$p(y, f, u) = p(f, u; X, Z)p(y|f) \quad (39)$$

where $p(f, u; X, Z) = p(u|Z)p(f|u; X, Z)$ and $p(u|z) = N(u|0, K(Z, Z))$. If $p(f|u)$ is known, the joint density of above can be obtained.

The relationship between u and f is given as follows since the u and f are generated by the same GP:

$$p(f|u; X, Z) = \mathcal{N}(f|\alpha(f), \beta(f)) \quad (40)$$

where the $\alpha(f)$ and $\beta(f)$ represent the mean and variance of f , respectively. They can be expressed as:

$$\alpha(f) = K(X, Z)k(Z, Z)^{-1}u \quad (41)$$

$$\beta(f) = K(X, X) - K(X, Z)K(Z, Z)^{-1}K(Z, X) \quad (42)$$

To obtain the true posterior $p(f, u)$ and lessen the computing overhead, variational inference is implemented. Its goal is to minimize the Kullback-Leibler divergence between the true posterior $p(f, u)$ and the variational posterior q in order to get an approximation posterior $q(f, u) = p(f|u)q(u)$. The minimization Kullback-Leibler divergence is equivalent to maximizing the evidence lower bound (ELBO) [44]:

$$\mathbb{L} = \log p(y) \geq \mathbb{E}_{q(f, u)} \log \frac{p(y, f, u)}{q(u, f)} \quad (43)$$

Further, the ELBO can be simplified as [44]:

$$\mathbb{L} \geq \mathbb{E}_{q(f)} \log p(y|f) - \text{KL}(q(u)||p(u)) \quad (44)$$

Therein, the marginal likelihood $q(f)$ is:

$$q(f) = \int p(f|u)q(u)du = \mathcal{N}(f|\mu^l, \Sigma^l) \quad (45)$$

where $q(u) = \mathcal{N}(\mu^l, \Sigma^l)$. The μ^l and Σ^l is expressed as:

$$\mu^l = K(X, Z)K(Z, Z)^{-1}u \quad (46)$$

$$\Sigma^l = K(X, X) + K(X, Z)K(Z, Z)^{-1}\left(\sum -K(Z, Z)\right)K(Z, Z)^{-1}K(Z, X) \quad (47)$$

These allow for the knowledge of every component of the simplified ELBO presented in Eq. (44).

Summarized, the parameters of SVGP that need to be optimized are $\Theta = \{\sigma_n, \sigma_k, \mu^l, \Sigma^l, Z\}$. They are optimized by maximizing the Eq. (44).

After the optimized parameters are obtained by maximizing the ELBO, input a new test point x^* in the test stage, and the posterior distribution of the predicted value can be calculated:

$$q(f^*) = \int p(f^*|u)q(u)du \quad (48)$$

The heat demand supplied by CHP, the voltages of each node in the MMG system, and the exchange power between the MMG system and DN can be obtained based on the distribution calculated by Eq. (48).

3.2. Decision-making approach based on MADRL

The thermal and electrical surrogate models eliminate the need of precise model parameters for the electricity network and heat network. This means that the calculation of objective function Eq. (23) can get rid of the model-based power and thermal flow calculation processes. Later, the thermal and electrical surrogate models integrate into the environment of the proposed MADRL approach, consisting the whole proposed physics-model-free control framework.

The optimization problem is reformulated as a Markov game, which

is solved by the proposed MADRL approach. There are several parts to model in the Markov game.

1) Problem reformulation

- **State:** State represent the current operating characteristics of the individual MG system and are collected by each MG controller and used to make energy management decisions. The state of the g -th MG at time t is defined as:

$$s_t^g = (t, \widetilde{\mathcal{P}}_t, \widetilde{SOC}_t^g, \widetilde{TL}_t^g, \widetilde{EL}_t^g, \widetilde{P}_{i,t-1}^{g,DEG}, \widetilde{P}_{i,t}^{g,WT}, \widetilde{P}_{i,t}^{g,PV}, \widetilde{NV}_t^g) \quad (49)$$

where t denotes the time information; $\widetilde{\mathcal{P}}_t$ is the forecasted electricity price of current time t ; $\widetilde{SOC}_t^g$ is the SOC of energy storage system within g -th MG at node i ; $\widetilde{TL}_t^g$ is the forecasted cumulative thermal load of each node $\widetilde{P}_{i,t}^{g,TL}$; $\widetilde{EL}_t^g$ is the forecasted cumulative electrical load of each node $\widetilde{P}_{i,t}^{g,EL}$; $\widetilde{P}_{i,t-1}^{g,DEG}$ is the working power of the DEG at time $t-1$; $\widetilde{P}_{i,t}^{g,WT}$ and $\widetilde{P}_{i,t}^{g,PV}$ is the power of forecasted WT and PV, respectively, and $\widetilde{NV}_t^g$ is the voltage of each node.

- **Action:** The action of g -th MG is defined as

$$a_t^g = (\widetilde{P}_{i,t}^{g,ESS}, \widetilde{P}_{i,t}^{g,DEG}, \widetilde{P}_{i,t}^{g,PtH}, \widetilde{Q}_{i,t}^{g,PV}, \widetilde{Q}_{i,t}^{g,WT}, \widetilde{Q}_{i,t}^{g,ESS}, \widetilde{Q}_{i,t}^{g,DEG}) \quad (50)$$

where $\widetilde{P}_{i,t}^{g,ESS}$, $\widetilde{P}_{i,t}^{g,DEG}$ and $\widetilde{P}_{i,t}^{g,PtH}$ denotes the active power of ESS, DEG and PtH unit, respectively; $\widetilde{Q}_{i,t}^{g,PV}$, $\widetilde{Q}_{i,t}^{g,WT}$, $\widetilde{Q}_{i,t}^{g,ESS}$ and $\widetilde{Q}_{i,t}^{g,DEG}$ is the reactive power of PV, ESS, DEG, WT, respectively.

- **Reward:** The reward function can be defined as:

$$r_t^g = - \left(\frac{(C_t^{EP} + C_t^{Voltage})}{G} + C_t^{g,DEG} + C_t^{g,Gas} \right) \quad (51)$$

where $\sum_t^T \sum_g r_t^g$ denotes the objective function of this paper.

- **Transition function:** The transition function $s_{t+1}^g = \mathcal{T}(s_t^g, a_t^g, \Xi)$ maps the current state s_t^g to the next state s_{t+1}^g based on the action a_t^g and the environment's unpredictability Ξ . The deterministic elements of the s_t^g are the ones most affected by action, as shown by the equation $SOC_{t+1}^g = SOC_t^g + a_t^g \Delta t / E_{max}^g$, where E_{max}^g is the ESS capacity of the g -th MG. The $\widetilde{\mathcal{P}}_t$, $\widetilde{TL}_t^g$, $\widetilde{EL}_t^g$, $\widetilde{P}_{i,t}^{g,WT}$, and $\widetilde{P}_{i,t}^{g,PV}$ in s_t^g depends on Ξ for their state transition, and the $\widetilde{NV}_t^g$ is influenced by both a_t^g and Ξ .

2) Proposed MADRL approach

The proposed MADRL approach consists of two neural network-based components: an actor part that responds to make decisions and a critic part that responds to help the actor part approximate the optimal control strategy.

Actor network: The state design of this paper Eq.(49) contains various TSD (e.g., $\widetilde{\mathcal{P}}_t$, $\widetilde{TL}_t^g$, $\widetilde{EL}_t^g$ etc.). These TSD are unaffected by action during the state transition process, but their trend with certain laws can influence decision-making. Therefore, the strategy's performance will be significantly impacted by TSD due to the lack of knowledge about their trend. If the historical trend characteristic of TSD can be captured, it can mitigate the problem of TSD data having a negative impact on decision-making [45]. From this point of view, the historical trend representation of current time TSD is critical for the agent to make decision. In addition, the actor part of the proposed approach only utilizes the local information to make decision. Agents can benefit from conditioning based on the entire history of their actions and states in such case [46]. The actor network of each agent uses the state of the current time and the action from the previous time as input into a recurrent neural

network (RNN)-like network (e.g., gated recurrent unit (GRU)) to get a representation of trajectory history information τ^g for g -th agent. Mathematically, the τ_t^g can be calculated as follows:

$$\tau_t^g = GRU(s_t^g, a_{t-1}^g) \quad (52)$$

where GRU denotes the GRU network, the technical details about it can be referred to [47].

Any increase in the number of agents in multi-agent systems will result in an exponential expansion of the action space. Environment with a large-scale action space requires DRL to deploy exploration mechanisms throughout its optimization process in order to avoid premature convergence. ASN technology is often used in conventional DRL to avoid the local optimum. For the instance, π denotes the actor network, the action can be mathematically describe as: $a_t^g = \pi(\tau_t^g) + \vartheta$ when taking into account the continuous action space impacted by the Gaussian noise ϑ . Instead of introducing noise into the action space, as is done in ASN, PSN does it into the actor network's parameter space, which can be expressed as: $a_t^g = \widehat{\pi}(\tau_t^g)$ where the π is affected by the PSN to get $\widehat{\pi}$. The actor network with ASN will cause independent of the input τ_t^g since, despite having a fixed input τ_t^g , the output a_t^g is different because of the ASN. Even though training can create a mapping relationship between input τ_t^g and output a_t^g , such a procedure is likely to weaken the connection between them in situations when ASN has a significant impact on actions, leading to subpar performance in complicated contexts [48]. The PSN is an effort to meet this demand. At the beginning of each episode, the PSN keeps itself fixed till the arrival of the next episode. Following a pattern like this ensures consistency in actions and introduces a direct dependency between the τ_t^g and the a_t^g [49]. A perturbed actor network (PAN) and an adaptive actor network (AAN) are used to implement PSN in actor networks. Importantly, PAN and AAN do not update parameters through gradients. The parameters of PAN and AAN networks are only updated once per episode in accordance with the actor network parameters and the sampled noise, and then remain fixed until the next episode rolls around. A linear layer of a neural network is used to characterize the PSN in detail, as shown below:

$$y = Wx + B \quad (53)$$

where $y \in \mathbb{R}^q$, $x \in \mathbb{R}^p$, $W \in \mathbb{R}^{q \times p}$ denotes the parameters, and $B \in \mathbb{R}^q$ represents the bias. The corresponding layer with PSN can be defined as:

$$y = (W + \vartheta^w)x + (B + \vartheta^b) \quad (54)$$

where the ϑ denotes the Gaussian noise variables, $\vartheta^w \in \mathbb{R}^{q \times p}$ and $\vartheta^b \in \mathbb{R}^q$.

In light of the fact that PSN's impact on outcomes is largely determined by the network structure, and that parameters' sensitivity to PSN changes throughout training, it is not recommended that PSN be sampled from a fixed noise distribution in actor networks [49]. In an alternative metric, the parameter space noise can be adaptively increased or decreased based on whether it falls below or above a threshold depending on the difference of output between the PAN and the actor network. This type of metric can be defined as follows:

$$\sigma_{t+1} = \begin{cases} \lambda \sigma_t, & \mathcal{D}(\pi, \widehat{\pi}) \leq \xi \\ \sigma_t / \lambda, & \mathcal{D}(\pi, \widehat{\pi}) > \xi \end{cases} \quad (55)$$

$$\mathcal{D}(\pi, \widehat{\pi}) = \sqrt{N^{-1} \cdot \sum_{i=1}^N \mathbb{E}_s \left[(\pi(\tau)_i - \widehat{\pi}(\tau)_i)^2 \right]} \quad (56)$$

where σ is the standard deviation of Gaussian noise; the λ is the scaling factor; the ξ is the threshold value and N represents the dimensions of action.

The actor network is utilized to establish the mapping relationship from the input state to the output decision making, and this process is crucial to the model's effectiveness. In order to map features from the input to the desired output, neural networks with multiple layers are

important because deep neural networks can improve the nonlinear mapping capability of actor network to extract features from the input and map them to the desired output. As the network becomes deeper, the original information from input data may gradually disappear, so simply increasing the depth of the network cannot be effective [50]. In order to maximize information retention and make greater advantage of the deeper neural network's non-linear mapping capabilities, D2RL handles the challenges deftly by linking the original input to each hidden layer. Mathematically, the mapping relationship from the g -th agent's input to the output using the D2RL neural network architecture can be defined as:

$$\begin{aligned} I_1 &= \tau^g, \\ I_{u+1} &= \text{ReLU}([I_u, \tau^g] \cdot W_u^g + B_u^g), \quad u = 1, \dots, U-1 \\ \zeta &= \text{ReLU}(I_U \cdot W_U^g + B_U^g), \\ a^g &= \tanh(\zeta \cdot W_O^g + B_O^g), \end{aligned} \quad (57)$$

where W_u^g and B_u^g are the weight and bias matrices of the u -th hidden layer of the g -th agent, respectively; $[I_u, \tau^g]$ represents the concatenation of I_u and τ^g , which is the critical operation of D2RL; U denotes the total number of hidden layers; ζ denotes the latent feature extracted by all the U hidden layers; and W_O^g and B_O^g are the weight and bias matrices of the output layer, respectively.

Summarized, the advantages of combining PSN and D2RL technologies are: The parameter-level operations of PSN and the complex and efficient parametric connection relationship built by D2RL introduce more exploratory activity than does the combination of shallow dense architecture and PSN. In addition, D2RL has better non-linear representational capability than shallow dense neural network architecture to fast track the link between input and optimal output decision making, thereby helping the MADRL approach surmount the instability and poor convergence speed of the training process brought on by the introduction of PSN.

Critic network: By utilizing the attention mechanism, the critic network effectively manages the MMG system's energy by coordinating strategies among actor networks..

The g -th agent action-value function Q_g can be obtained according to the following steps:

$$e_1 = \mathcal{F}_1(\tau^1, a^1), \dots, e_g = \mathcal{F}_g(\tau^g, a^g), \dots, e_G = \mathcal{F}_G(\tau^G, a^G) \quad (58)$$

$$[\alpha_{g(1)}, \dots, \alpha_{g(g-1)}, \alpha_{g(g+1)}, \dots, \alpha_{g(G)}] = \frac{f(e_g w_Q^h + b_Q^h, [e_1 w_K^h + b_K^h, \dots, e_G w_K^h + b_K^h])}{\sqrt{d_k}} \quad (59)$$

$$\omega_i = \frac{\exp(\alpha_{g(i)})}{\sum_{i \neq g} \exp(\alpha_{g(i)})} \quad (60)$$

$$O_h = \sum_{i \neq g} \omega_i (e_i w_V^h + b_V^h), \quad h \in [1, H] \quad (61)$$

$$Q_g = \mathcal{G}_g(e_g, O_1, \dots, O_h, \dots, O_H) \quad (62)$$

where e_1, e_g , and e_G represent the encoders of the (τ^1, a^1) , (τ^g, a^g) , and (τ^G, a^G) pairs, respectively; $\mathcal{F}_g$ denotes the feed forward neural network of the g -th agent with an input layer, two hidden layer of rectified linear units function neurons, and a linear output layer; $f(\mathcal{U}, [\mathcal{V}, \dots, \mathcal{W}]) = [\mathcal{U} \cdot \mathcal{V}, \dots, \mathcal{U} \cdot \mathcal{W}]$; d_k represents the network dimensions, and attention layer with H heads are used in the proposed approach; in each head, there are six optimizable parameters $w_Q^h, b_Q^h, w_K^h, b_K^h, w_V^h, b_V^h, h \in [1, H]$; $\mathcal{G}_g$ denotes the feed forward neural network similar to $\mathcal{F}_g$.

3.3. Training of the proposed physical-model-free control framework

To begin, the surrogate model is trained in a supervised manner. The proposed MADRL-based control approach is then combined with surrogate models. The ongoing interaction between the two can generate

desired control strategy. The flowchart of the training process are shown in Fig. 2. The following also includes a detailed description of training processes.

In the beginning of training, surrogate models are trained in a supervised manner by minimizing the following metric:

$$MAE = \frac{1}{N} \sum_{i=1}^N |f_{true} - f_{predict}| \quad (63)$$

where f_{true} and $f_{predict}$ denote the true and predicted value. When surrogate models finish their training, the parameters and memory of MADRL-based approach are initialized. After initializing, the training of MADRL begins and ends until the training episode is equal to M_1 . Each episode represents a random sample of a day of operation from a training set, in which T time-steps make up an operation day. In each time-step, the action of each agent is chosen at random to explore the action space. When the memory size exceeds M_2 , each agent performs its action in accordance with $a_t^g = \pi(\tau_t^g)$. When all agents execute their actions a_t^g , each one is immediately rewarded r_t^g , and the state of system s_t^g moves on to the next state s_{t+1}^g . Following that, each agent stores the experienced pairings $(s_t^g, a_t^g, r_t^g, s_{t+1}^g)$ in their memory. After storing the pairs, the parameters of the actor network can be updated in order for it to ascend the gradients shown below:

$$\nabla J = \mathbb{E} \left[\nabla \log(\pi^g(a_t^g | \tau_t^g)) \left(\begin{array}{c} Q^g(\Gamma_t, A_t) \\ -\alpha \log \pi^g(a_t^g | \tau_t^g) \end{array} \right) \right] \quad (64)$$

where $\Gamma_t = [\tau_t^1, \dots, \tau_t^G]$ and $A_t = [a_t^1, \dots, a_t^G]$; the use of temperature parameter α is intended to balance exploration and exploitation in the process of training. The parameters of critic network can be updated to minimize the following equation:

$$\mathcal{L} = \sum_{g=1}^G \mathbb{E} [(Q^g(\Gamma_t, A_t) - \mathcal{J}^g)^2] \quad (65)$$

$$\mathcal{J}^g = r_t^g + \gamma \mathbb{E} [Q_{tar}^g(\Gamma_t, A_t) - \alpha \log(\pi_{tar}^g(a_{t+1}^g | \tau_{t+1}^g))] \quad (66)$$

where γ is the reward discount factor; Q_{tar}^g and π_{tar}^g are the target critic and actor networks that are used to stabilize the training, whose parameters are updated according to the soft update mechanism [51]; $-\log(\pi_{tar}^g(a_{t+1}^g | \tau_{t+1}^g))$ indicates the entropy regularization. Once time step t equals T , it means that an episode is over, and the parameter noise within PAN and the noise distribution are updated for the preparation of a new episode.

4. Case studies

In this section, the detailed parameter settings are first provided, followed by the performance evaluation of the proposed physics-model-free control framework, which consists of surrogate model and MADRL-based control model.

4.1. Case study setup

Three MGs make up the MMG system in the case study. Each MG is represented by a modified IEEE 13-bus network [52] that includes a heat network and an electricity network. Fig. 3 shows the architecture of MMG system in case study. In the figure, each MG has ES, DE, WT, PV, CHP and PtH unit. The used datasets for electrical load, thermal load, electricity price, wind energy, and solar energy, respectively, are [24,53,54,55], and [56]. The 200-day training set (from January 1 to July 19) and the 100-day test set (Jul. 20 to Oct. 27) are created from these datasets. The related parameters of the optimization model are listed in Tables 2 and 3. The parameters of the proposed control framework are listed in the Table 4.

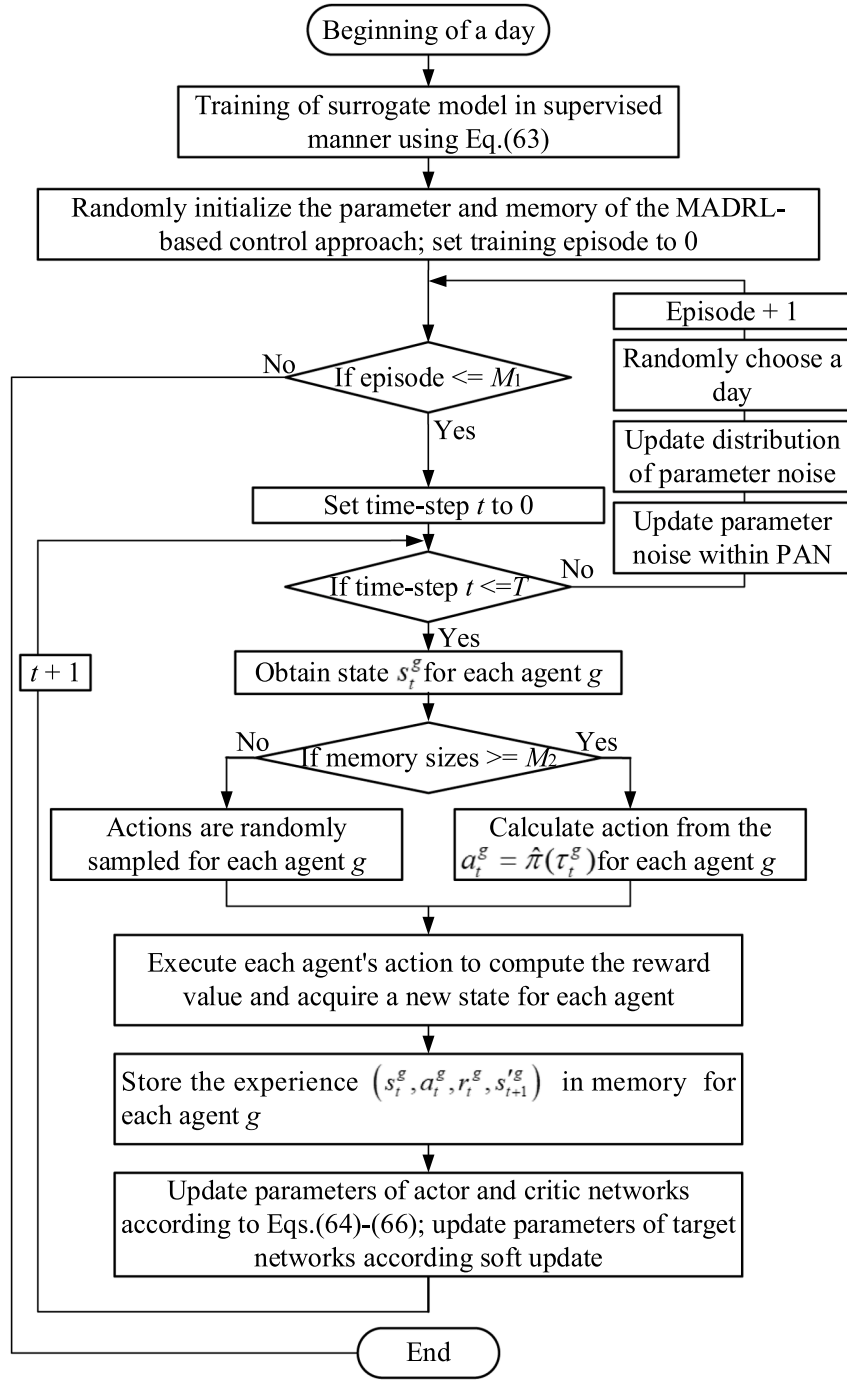


Fig. 2. Flowchart for the training of proposed physical-model-free control framework.

4.2. Evaluation of the proposed surrogate model

There are three surrogate models that need to be applied in the optimization of MMG system energy management: voltage, exchange power, and CHP-generated energy surrogate models, respectively. The mapping relationship of training of these three surrogate models are summarized in Table 5. To assess the effectiveness of the proposed surrogated model, a comparison is conducted between linear regression (LR), deep neural networks (DNN), and the proposed SVGP as surrogate models to evaluate their fit accuracy when there are limited data points available. Therein, DNN contains two hidden layers with neuron counts of 100 and 100, respectively. 32 and 0.001 are chosen as the batch size and learning rate. 2000 episodes are used to train the DNN. Tables 6, 7,

and 8 summarize the comparison results for the voltage, exchange power, and CHP-generated energy surrogate models, respectively, using MAE as the metric. As the Tables 6, 7, and 8 show, when the limited instances, i.e., 20, 40, and 80 instances of training data are used to train the surrogate model, the proposed SVGP-based approach can achieve a lower MAE than that obtained by the LR and DNN-based surrogate models. Therein, the LR performance shown in Table 6 has the poorest performance (SVGP outperformed it by 98.4% at 20 instances), the DNN performance shown in Table 7 has the poorest performance (SVGP outperformed it by 93.6% at 80 instances), and all three approaches can achieve similar performance in the CHP-generated energy surrogate model shown in the Table 8. In the complex mapping relationship surrogate model, i.e., the voltage and power exchange surrogate model, the

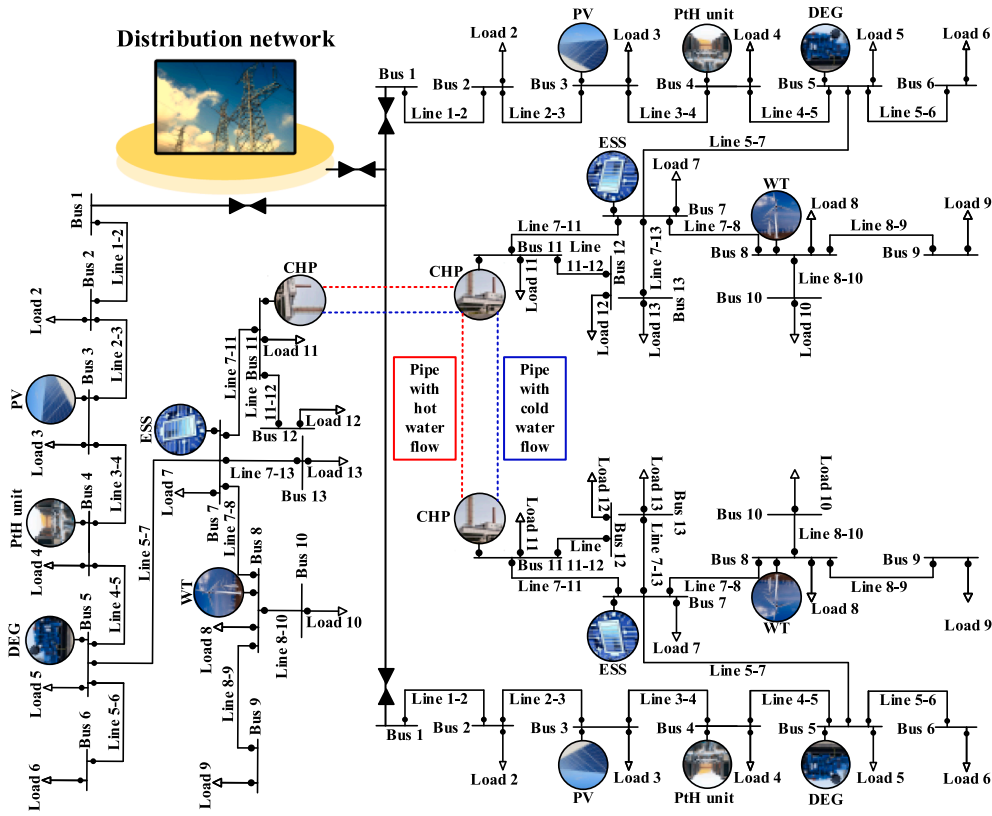


Fig. 3. The MMG system in case study.

Table 2
Model Parameters in Section II.

Symbol	Value	Symbol	Value
a (\$/(kW) ² h)	1040	$C^{start/stop}$ (\$)	5
b (\$/MWh)	30.4	η_{CHP}	0.36
c (\$)	1.3	L_{cvmg} (MWh/m ³)	9.7×10^{-3}

Table 3
Parameters of the MMG System.

	P_{PV}^{max}	P_{WT}^{max}	P_{PtH}^{max}	P_{DEG}^{max}	P_{DEG}^{min}
MG1	0.2 MW	0.25 MW	0.005 MW	0.05 MW	0.02 MW
MG2	0.15 MW	0.2 MW	0.005 MW	0.05 MW	0.02 MW
MG3	0.1 MW	0.15 MW	0.005 MW	0.05 MW	0.02 MW
	P_{ES}^{max}	E_{max}	SOC_{ES}^{max}	SOC_{ES}^{min}	$R_{up/dn}^{DEG}$
MG1	0.05 MW	0.2 MWh	1	0.1	0.025
MG2	0.045 MW	0.18 MWh	1	0.1	0.025
MG3	0.04 MW	0.16 MWh	1	0.1	0.025

Table 4
Parameters of the Proposed Control Framework.

Parameters	Value
Threshold value	0.1
Scaling factor	1.001
Temperature parameter	0.1
Reward discount factor	0.95
Memory capacity	1e6
Learning rate of actor	3e-4
Learning rate of critic	3e-4
Soft replacement	1e-3
Batch sizes for updating	256
Optimizer	Adam

LR and DNN cannot do a good job of taking into account different complex mapping relationships at the same time. This means that the LR and DNN-based surrogate models are difficult to achieve the desired results in the different flow calculations, but the proposed SVGP-based surrogate model can achieve that. The above results demonstrate the effectiveness of the proposed SVGP-based surrogate model.

4.3. Evaluation of the proposed physics-model-free-based control framework

We evaluate the effectiveness of the proposed control framework against several baselines.

a) Descriptions of baselines:

- TD3 [57]: The MG controllers are modeled as TD3 agents, with each one being trained to maximize its own reward function while disregarding procedures for coordinating with other MGs.
- MATD3 [30]: MATD3 is an optimization technique described in [30] to optimize energy management in a MMG system with integrated energy.
- MASAC [33]: The MASAC approach is described in [33], which aims to improve the economic dispatch and stability of the MMG system by optimizing energy management.
- MAAC [58]: MAAC is a MADRL approach that integrates the attention layer to enhance agents' coordination.
- Proposed MADRL approach: As opposed to the proposed physics-model-free control framework that utilizes surrogate models to obtain power and thermal flow results, this approach utilizes mathematical power and thermal flow model to obtain calculation results. Setting this baseline will allow us to evaluate the effectiveness of the surrogate model proposed.
- Optimization of deterministic information scenario with NLOpt [17]: NLOpt's optimization setting is based on perfect future information. The perfect future information means that the future electricity

Table 5
Mapping Relationships of Different Surrogate Models.

	Input	Output
Voltage	$TL_t^g, EL_t^g, P_{i,t-1}^{g,DEG}, P_{i,t}^{g,WT}, P_{i,t}^{g,PV}, NV_t^g, P_{i,t}^{g,ESS}, P_{i,t}^{g,DEG}, P_{i,t}^{g,PH}, Q_{i,t}^{g,PV}, Q_{i,t}^{g,WT}, Q_{i,t}^{g,ESS}, Q_{i,t}^{g,DEG}$	NV_{t+1}^g
Exchange power	$TL_t^g, EL_t^g, P_{i,t-1}^{g,DEG}, P_{i,t}^{g,WT}, P_{i,t}^{g,PV}, NV_t^g, P_{i,t}^{g,TL}, P_{i,t}^{g,EL}, P_{i,t}^{g,ESS}, P_{i,t}^{g,DEG}, P_{i,t}^{g,PH}, Q_{i,t}^{g,PV}, Q_{i,t}^{g,WT}, Q_{i,t}^{g,ESS}, Q_{i,t}^{g,DEG}$	p_{t+1}^{DN}
CHP	$P_{i,t}^{g,TL}$	$P_{i,t+1}^{g,CHP}$

Table 6
MAE of Voltage Surrogate Models with Different Numbers of Training Instances.

Instances	LR	BPNN	SVGP
20	0.2923	0.0703	0.0048
40	0.1598	0.0181	0.0028
80	0.1390	0.0133	0.0026

Table 7
MAE of Exchange Power Surrogate Models with Different Numbers of Training Instances.

Instances	LR	BPNN	SVGP
20	0.0608	0.3503	0.0459
40	0.0536	0.2797	0.0281
80	0.0213	0.1539	0.0098

Table 8
MAE of CHP Generated Energy Surrogate Models with Different Numbers of Training Instances.

Instances	LR	BPNN	SVGP
20	1.570e ⁻⁴	1.755e ⁻⁴	1.310e ⁻⁴
40	1.400e ⁻⁴	1.328e ⁻⁴	1.152e ⁻⁴
80	1.233e ⁻⁴	1.205e ⁻⁴	1.145e ⁻⁴

price, electrical load, etc. can be known beforehand, i.e., NLOpt yields optimized results based on deterministic and accurate information.

b) Performances on the test set:

The convergence processes of the proposed and other MADRL approaches are shown in Fig. 4, in which we evaluate the performance on the test set one time every 10 episodes of parameter updates, i.e., there are 30,000 episodes of training and thus 3000 evaluations on the test set. In comparison to other approaches, the reward curve for TD3 shows the lowest value and oscillates greatly during training, demonstrating that such a lack of coordination is not appropriate for energy management in MMG system. The MATD3 proposed in [30] is an improvement over the TD3 in that it use the multi-agent framework to build the coordination

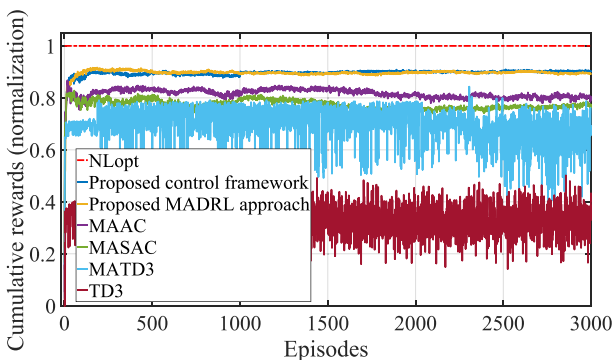


Fig. 4. The convergence process of proposed control framework and baselines on the test set.

relationship between each MG for the energy management in an MMG system. There is little doubt that MATD3's coordinated mechanism gives it a performance and stability edge over TD3. The MADRL with the entropy-based approach, MASAC, can obtain better performance than the MATD3 approach. To improve upon the MASAC's already impressive performance, the authors of [58] presented the MAAC, which use the attention layer in the critic network to significantly improve coordination between MGs. Setting the attention network in MAAC is meant to speed up the building of consensus among the agents and improve their coordination by including some of the network parameters shared across the agents in each agent's own neural network parameters. Furthermore, the proposed MADRL approach first uses GRU to generate historical information representation in order to overcome the negative impact of the TSD on the decision-making process. Additionally, due to the exponential increase in action space in multi-agent environments, premature convergence to local optimal solutions is much more likely than in single-agent environments. The proposed MADRL approach introduce PSN technology into the neural network to overcome the negative influence bring by the local optimal solutions. Then, proposed MADRL approach leverages D2RL to get rid of the noise-related instability and inefficiency of PSN exploration while also providing richer parameter connection relationships to construct a more powerful exploration mechanism overall. The proposed MADRL approach achieves better performance and stability than MAAC through an efficient exploration mechanism incorporating historical information representation. The proposed MADRL approach relies on the accurate physical information of the MMG system and uses mathematical power and thermal flow model to calculate power and thermal flow results, whereas the proposed control framework uses surrogate models to acquire results for power and thermal flow. This eliminates the need for precise model parameters of electricity network and heat network, which are notoriously hard to come by and are subject to a wide range of practical errors. Although these two ways of calculating power and thermal flow are different, the proposed physics-model-free control framework has similar performance to the proposed MADRL approach. As a result, the surrogate model can provide a precise reward signal for the training of DRL agents, thus demonstrating the effectiveness of the proposed control framework. In order for the energy management of MMG to be stated as a deterministic optimization problem and addressed by NLOpt, we assume that the unknown variables are known in advance. The research objectives of this paper include two parts: one is economic cost, and the other is voltage stability. NLOpt is a traditional model-based centralized optimization algorithm based on perfect future information that outperforms the proposed control framework. NLOpt spends 8.95% less on the economic cost compared to the proposed control framework, 38.27% less on the voltage cost compared to the proposed control framework, and 13.80% less on the total cost compared to the proposed control framework. However, because there is unpredictability, it cannot be used in a realistic context. The proposed control framework not only gets rid of the premise that one has accurate knowledge of the parameters and topology of the electricity and heat networks but also has the closest performance to that of NLOpt compared with other baselines. The results further demonstrate the effectiveness of the proposed control framework.

c) Optimizing stability:

One phenomenon is that, despite having the same number of updates each time, a DRL-based approach's final performance will not be the same in these times. In other words, the excellent performance of a DRL-based approach may be accidental. Due to the nature of DRL optimization, there is sometimes a situation where it is difficult to escape after falling into a local optimal trap. In the process of agent falling into the trap for a long time, gradient information that may have a persistent negative impact on the update of parameters will accumulate. The negative impacts brought about by gradient information in parameter updates are difficult to erase, even when new gradient information that may have a positive impact on parameter updates is introduced. Similar deceptive traps are very common in complex problem optimization, which is why the reward curve changes from stable convergence to difficult convergence when DRL with poor optimization ability is applied in environments with increasing optimization difficulty [59]. The research of this paper involves internal multi-energy allocations within individual MGs, external multilateral multi-energy interactions among interconnected MMGs, and energy trading between MMGs and the distribution network. In this context, any decision will have multiple impacts for the MMG system, which increases the difficulty of the agent's seeking optimization in the decision space. Thus, the above mentioned phenomenon is more serious in the considered multi-agent environments of this paper. In order to verify the performance stability of the proposed control framework, box plots of proposed control framework and baselines are provided in Fig. 5. As the figure shows, the TD3 has the worst stability due to its lack of coordination. With the introduction of a coordinated mechanism, the MATD3 is capable of achieving better and more stable performance than the TD3. The experiment results show that the multi-agent control problem in the coordinated environment needs to introduce the mechanism of coordinated agent interaction into the control algorithm. The advantage of this is that it can avoid the negative impact on the parameter update of the control algorithm caused by falling into the local optimal trap frequently and for a long time. Further, the MASAC approach can have better performance than the MATD3 approach due to the introduction of entropy. Through the comparison of the performance of MASAC and MAAC, it is known that the use of attention mechanism can enhance the coordinated relationships between agents, thus reducing the accumulation of negative gradient information caused by falling into the local optimal trap, and further improving the performance and stability of the control algorithm. Furthermore, the proposed MADRL approach and physics-model-free control framework not only achieve the lowest cost but also have the most stable performances among various approaches, which demonstrating that (1) The desired results achieved by the proposed MADRL approach are not accidental. The efficiently designed exploration mechanism incorporating historical information representation can ensure the MADRL approach gets rid of the deceptive trap as much as possible. Thus, the stability of the desired performance achieved by the proposed MADRL approach can be guaranteed; (2) In the training of DRL agents, the surrogate model can provide a precise

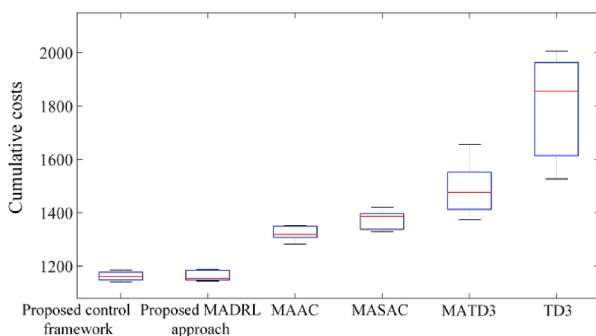


Fig. 5. Box plots of the performance of various approaches under the 100-day test set.

reward signal for the agents, thus resulting in the proposed control framework having similar performance and stability to the proposed MADRL approach that relies on physics models. It is evident from this that the proposed control framework is effective.

d) Solution quality:

The Fig. 6(a) shows the real-time electricity price for one day. The Fig. 6(b) shows the voltage profile of the MMG system under control, in which the red tone, green tone, and blue tone lines denote the voltages of MG1, 2, and 3, respectively. The gradient color in the same tone represents the voltage curve of different nodes in the same MG. The Fig. 6(c) shows the charging (positive value)/discharging (negative value) power of ESSs of different MGs. The ESS can leverage the temporal differences in electricity prices to obtain energy arbitrage and utilize charge/discharge behavior to maintain voltage stability. (1) When the electricity prices are low, ESS2 and 3 execute charging behavior for the energy arbitrage objective at the intervals of 4:00, 5:00. (2) When the voltage are high, ESS1 utilizes charge behavior to store the rich electric energy in the grid to reduce the high voltage at the intervals of 7:00 ~ 10:00, in which charging power gradually decreases with the rise in electricity prices, considering the energy arbitrage target. (3) When the electricity prices are low and the voltages are high, the ESS1-3 have charging behavior at intervals of 21:00 ~ 23:00 in order to suppress voltage climbing while also purchasing electrical energy storage at a low cost, in which the charging behavior of ESS considers the economic cost at the same time, and as such, as the electricity price decreases (21:00 ~ 23:00), the charging power will gradually increases (21:00 ~ 23:00). (4) When the electricity prices are high and the voltages are low, ESS1 discharges energy at the intervals of 11:00 ~ 19:00, and ESS2 and 3 discharges energy at the intervals of 9:00 ~ 19:00.

Fig. 6(d) depicts the operational status of the PtH units. This unit transforms electrical energy to heat energy. As the PtH is also a component that consumes power, much like the ESS, its power profile follows the similar rules to that of the ESS. CHP and PtH units supply the thermal demand inside the heat network. When electricity prices are low (i.e. between 0:00 and 7:00, 22:00 and 23:00), thermal energy is mostly supported by the PtC using electrical energy. When electricity prices are high (i.e. between 8:00 and 21:00), thermal energy is predominantly supported by the CHP.

Fig. 6(e) shows the DEG working situation. The DEG1, 2, and 3 operate at minimum power at intervals 0:00 ~ 8:00. This is due to the fact that the electricity prices are low at these intervals, so purchasing electricity directly from the DN to satisfy demand is more cost effective. In addition, from 0:00 to 8:00, the voltage of MG1, 2, and 3 is relatively stable. To reduce the DEG operation and fuel costs, it is reasonable that DEG operates in the minimum power when facing a more stable voltage. After 8:00, as people gradually start to get busy, the daily electricity consumption starts to increase. This leads to the electricity price starting to rise and the voltage starting to fluctuate significantly, which requires DEG to work to a certain extent to reduce the electricity purchased from the DN for the cost-effective objective and supply the electrical energy for the voltage-stable objective. For example, the DEG2 and 3 operate at almost maximum power from 9:00 to 19:00. This is due to the fact that (1) there are high electricity prices and (2) MG2 and MG3 have the most severe undervoltage conditions during that time period. DEG1's working power is similar to DEG2 and DEG3, but the phenomenon of DEG1's power changing with voltage is more pronounced due to MG1's greater voltage stability than MG2 and MG3 (e.g., during 12:00 ~ 20:00, when the voltage decreases, the power of DEG1 gradually increases to achieve the purpose of raising the voltage, and when the voltage rises, the power of DEG1 gradually decreases to slow down the trend of voltage continuing to rise).

e) Ablation study:

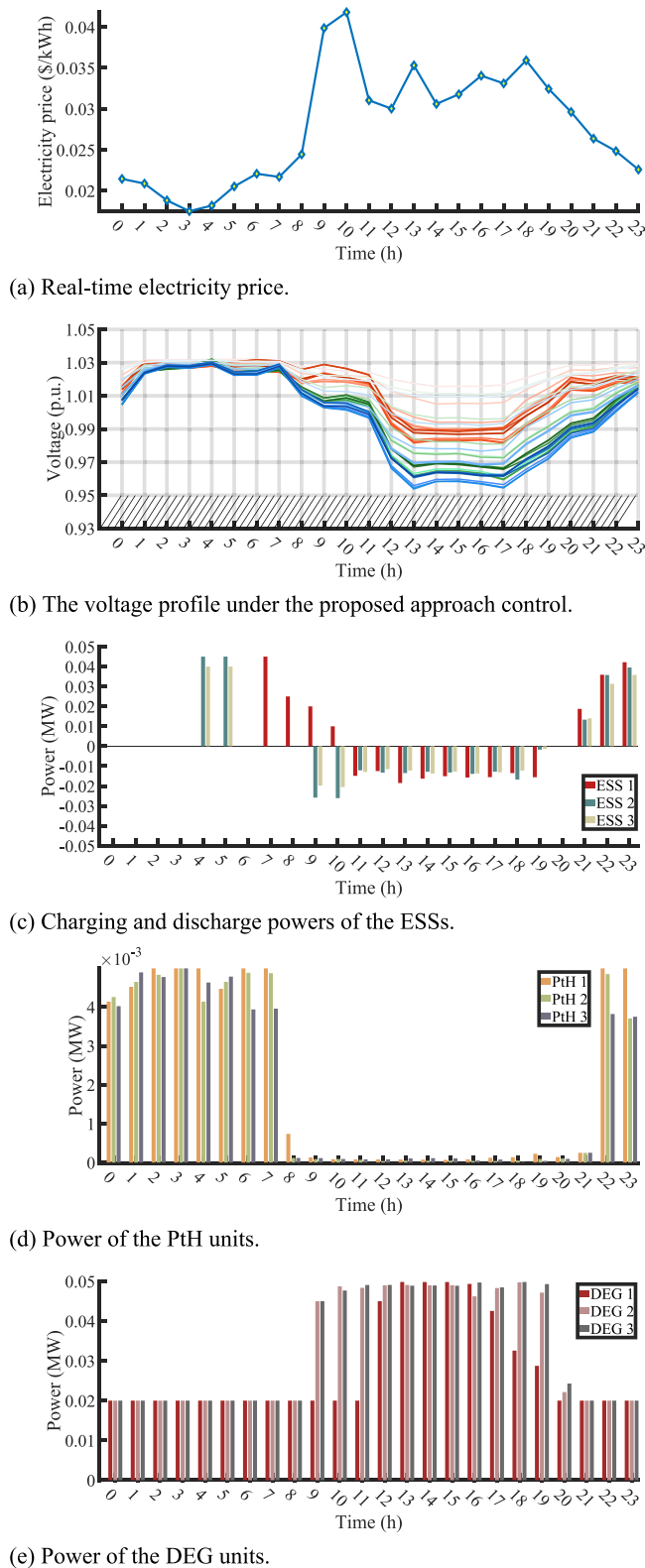


Fig. 6. Energy management results of the proposed control framework on a test day.

In order to further demonstrate the effectiveness of proposed control framework, the ablation study is given. Table 9 summarizes the different combinations. The proposed physics-model-free control framework, i.e., M1 is consisted of BasicMADRL, GRU, PSN, D2RL, and SurrogateModel. The other three combinations, M2, M3 and M4, have one less component

Table 9

Different Components for Ablation Study.

	Components
M1	BasicMADRL + GRU + PSN + D2RL + SurrogateModel
M2	BasicMADRL + GRU + PSN + D2RL + SurrogateModel
M3	BasicMADRL + GRU + PSN + D2RL + SurrogateModel
M4	BasicMADRL + GRU + PSN + D2RL + SurrogateModel

each than M1. The convergence processes of the four combinations are shown in Fig. 7, in which we evaluate the performance on the test set one time every 10 episodes of parameter updates, i.e., there are 30,000 episodes of training and thus 3000 evaluations on the test set. Obviously, the M1, i.e., proposed control framework has the best performance compared with others. This is due to the introduction of GRU and the exploratory mechanisms consisting of D2RL and PSN. M2 removes the GRU compared with M1 and obtains worse performance compared with M1. This is due to the TSD having a negative impact on the M2 and influencing its decision-making. The simulation results demonstrate that if the state contains TSD, their trend with certain laws can have a negative influence on the decision-making process, and GRU can mitigate the negative impact. M3 removes the D2RL, resulting in worse performance than M1. Additionally, the reward curve of M3 exhibits the largest fluctuation in the process of convergence compared with other approaches. The simulation results demonstrate that, compared with conventional parametric link relationships [45,51], D2RL establishes a more complicated and effective parametric link relationship that is more suitable to combine with PSN. This increases the potential for exploratory action while making sure that training is stable and efficiency. There is a decrease in performance in M4 due to the removal of the PSN. Based on the simulation results, it is evident that only applying GRU and D2RL cannot achieve the desired results in the multi-agent scenario due to the large action space that needs to be optimized, and as such, the effective exploratory mechanism is critical for the optimization of the multi-agent scenario.

Using the results of the ablation study, the proposed control framework is analyzed to show why it performs well. The experimental results show that by using a representation of historical information to counteract the bad effects of TSD and an effective exploration mechanism to find the desired solution, the proposed control framework solves the given control problem and yields good performance gains.

5. Conclusions

In this study, the proposed control framework manages the heat-electricity energy MMG system that both considers the economic and safe operation objective. The proposed control framework is consisted of surrogate model and MADRL approach. The surrogate model is introduced to imitate the realistic power and thermal flow to get rid of the assumption of one has accurate knowledge of the parameters and topology of the electricity and heat networks. A Bayesian theory-based

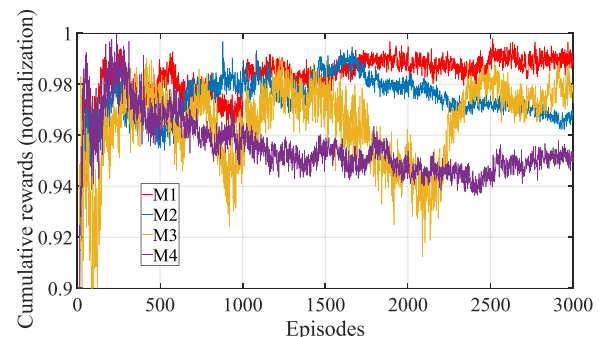


Fig. 7. The convergence process of M1, M2, M3, and M4 on the test set.

technique is used to ensure that a surrogate model can successfully identify the mapping connection between input and output using a minimum amount of historical information. The MADRL approach makes use of historical information representation to overcome the negative impact brought by TSD and an efficient exploration mechanism to seek optimization. The trained surrogate models are integrated into the MADRL environment during the training period of MADRL, allowing the MADRL approach to develop a desired energy management strategy based on the continuous interaction with the surrogate models. The simulations and comparative results reveal the following: 1) The proposed MADRL approach and control framework are the most stable, and their performance is closest to that of the NLOpt centralized approach based on deterministic global information, among the RL-based approaches. Moreover, due to the surrogate model's capacity to generate an accurate reward signal for the agents, the proposed control framework is as effective and reliable as the proposed MADRL approach, which uses physics models. 2) From the results of box plots, we can know that the excellent performances of proposed control framework is not accidental. 3) The interpretability of detailed performances in the MMG system further demonstrate the effectiveness of the proposed control framework. 4) The proposed control framework is rigorously evaluated based on the ablation study, proving its efficacy.

CRediT authorship contribution statement

Sichen Li: Conceptualization, Investigation, Methodology, Software, Validation, Visualization, Writing – original draft. **Weihao Hu:** Formal analysis, Supervision, Writing – review & editing. **Di Cao:** Formal analysis, Investigation, Validation, Writing – review & editing. **Zhe Chen:** Formal analysis, Supervision, Validation, Writing – review & editing. **Qi Huang:** Resources, Writing – review & editing. **Frede Blaabjerg:** Supervision, Validation, Writing – review & editing. **Kaiji Liao:** Investigation, Resources, Software.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Please refer to the references for the data.

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